

SYSTEM.ONCO

Explainable 3D Brain MRI Research Workspace for Volumetric Brain-Tumor Segmentation

FROM VOLUME TO INSIGHT

A unified research workspace for 3D brain MRI segmentation that combines deep learning with interpretability to enable transparent, analyzable and reproducible tumor analysis.



MULTI-MODAL MRI INPUT

T1, T1c, T2, FLAIR



3D CNN SEGMENTATION

Volumetric Analysis



EXPLAINABILITY (Grad-CAM)

Understand Model Focus



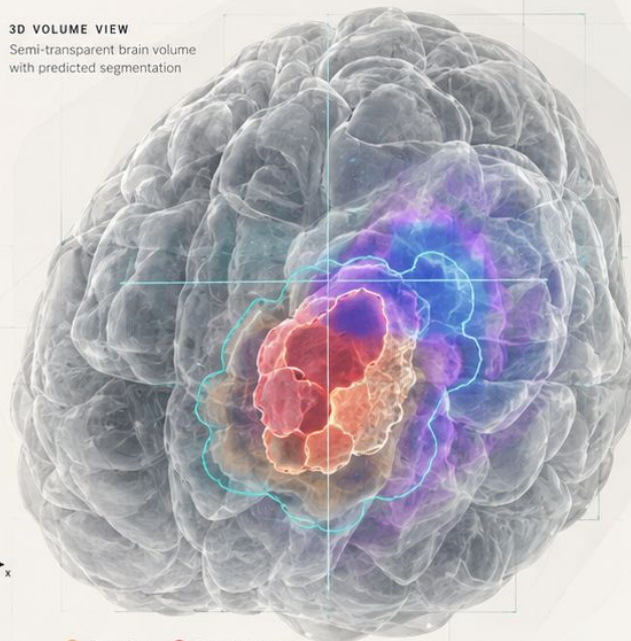
INTERACTIVE VISUALIZATION

Explore · Compare Analyze

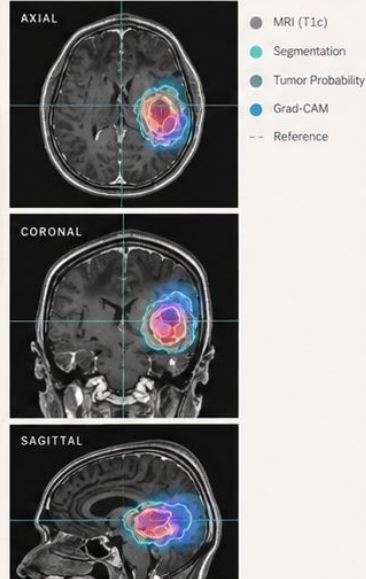


- Whole Tumor
- Tumor Core
- Enhancing Tumor
- Tumor Probability
- Grad-CAM (Model Focus)
- Reference Annotation

3D VOLUME VIEW
Semi-transparent brain volume with predicted segmentation



ORTHOGONAL MRI SLICES (WITH OVERLAYS)

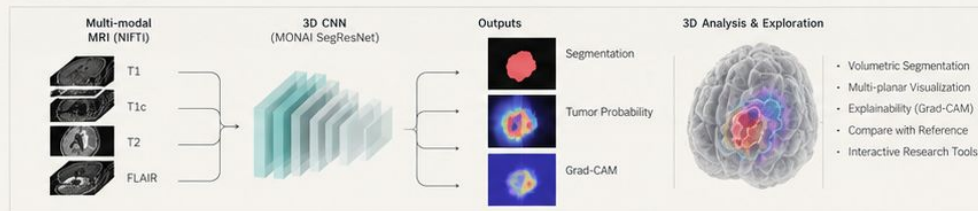


MULTI-PLANAR PERSPECTIVE

Consistent 3D segmentation across all views

MORE THAN A SEGMENTATION.
A DEEPER UNDERSTANDING.

WORKFLOW OVERVIEW



SAME
DATA.
DEEPER
QUESTIONS.
BRIGHTER
INSIGHTS.

01

THE
CHALLENGE

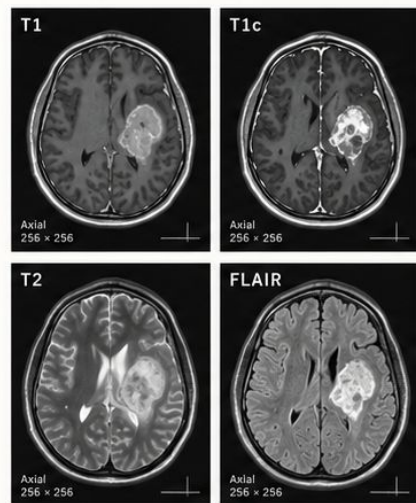
FROM MRI VOLUME TO EVIDENCE.

An explainable research workspace for 3D brain-tumor segmentation.

A VOLUMETRIC PERSPECTIVE FOR A DEEPER UNDERSTANDING.

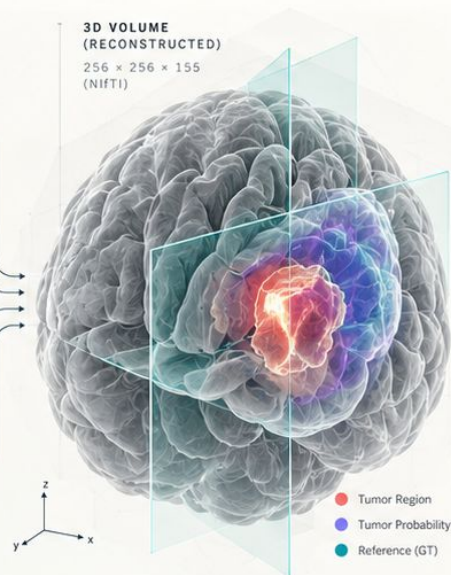
MRI
AI
VISUALIZATION
EXPLAINABILITY
RESEARCH

MULTI-MODAL BRAIN MRI (SAME SUBJECT)



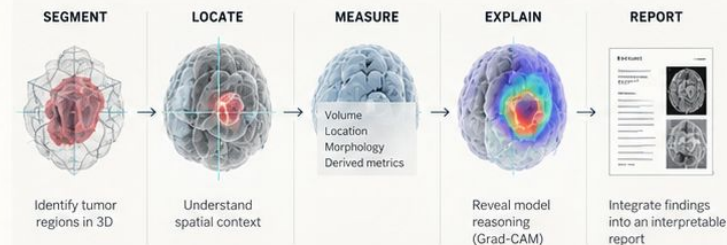
T1 | T1c | T2 | FLAIR

COMPLEMENTARY INFORMATION. A RICHER PICTURE.



FROM SLICES TO A UNIFIED 3D UNDERSTANDING

FROM DATA TO INSIGHT



KEY CHALLENGES

VOLUMETRIC
COMPLEXITY

A brain MRI is a 3D, multi-modal study with rich but heterogeneous information.

MODEL OUTPUT ≠
UNDERSTANDING

A segmentation mask alone does not explain what the model learned or why.

RESULTS NEED
CONTEXT

Quantitative outputs must be interpreted in anatomical, clinical, and research context to be meaningful.



Goal: connect segmentation, visualization, explanation, measurement, and reporting in one workflow.

/ TOWARDS
MORE
TRANSPARENT
NEURO-ONCOLOGY

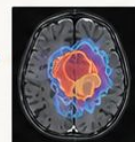
SYSTEM.ONCO: ONE STUDY, MULTIPLE LAYERS OF EVIDENCE.

INTERACTIVE VISUALIZATION. TRANSPARENT ANALYSIS. REPRODUCIBLE INSIGHTS.



EVIDENCE LAYERS

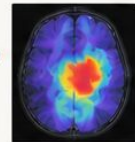
BEYOND A SINGLE PREDICTION



1. CNN SEGMENTATION

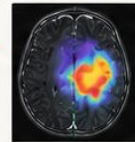
Tumor Core - Whole Tumor - Enhancing Tumor

- Enhancing Tumor
- Tumor Core
- Whole Tumor

STRUCTURED
PREDICTION OF
TUMOR REGIONS

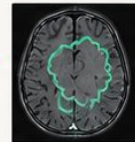
2. PROBABILITY

Voxel-wise sigmoid probability

SPATIAL
UNCERTAINTY
IN MODEL
PREDICTIONS

3. REGIONAL GRAD-CAM

Positive attribution around predicted tumor core

EXPLAINS
WHAT THE MODEL
FOCUSES ON

4. REFERENCE MASK

Dataset annotation - separate from inference

- Reference Annotation
- Model Segmentation (for comparison)

GROUND TRUTH
FOR VALIDATION
AND ANALYSIS

FROM VISUALIZATION TO INSIGHT



MEASURE

Quantify tumor region
and derived metrics

LOCATE

Explore in 2D and 3D
across modalities

COMPARE

Analyze prediction, probability,
Grad-CAM and reference

REPORT

Generate interpretable
findings and visual evidence

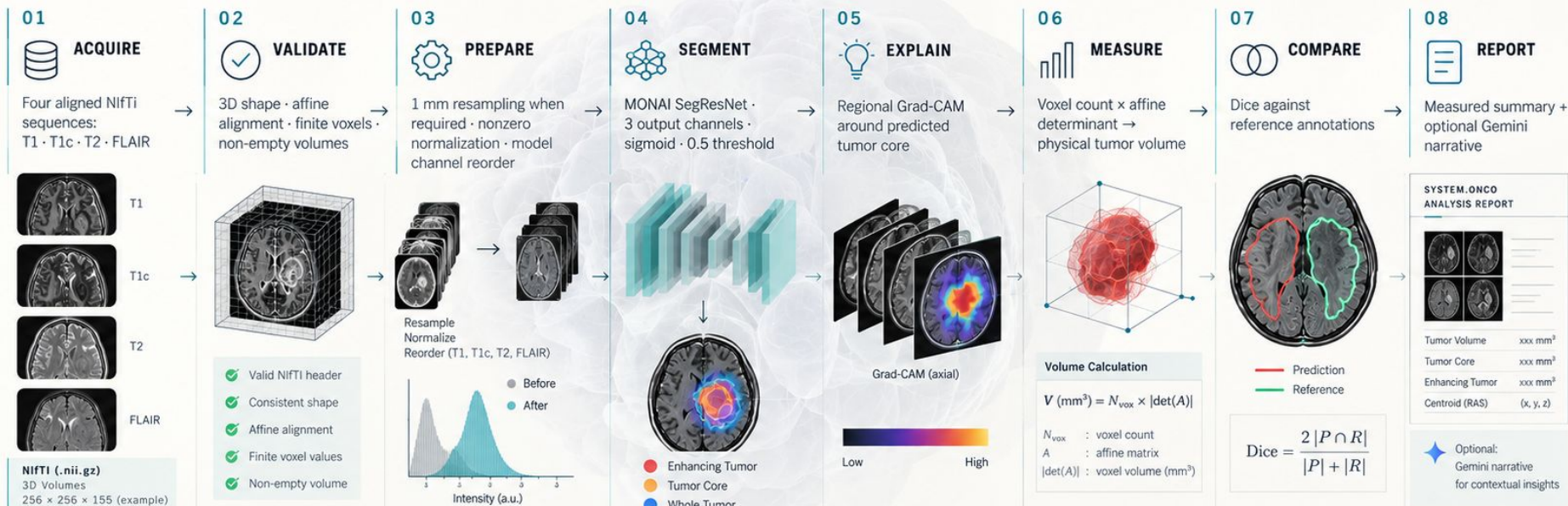
Reference masks never enter inference.

They are used only for evaluation, comparison and analysis.

UNDER THE HOOD: FROM NIFTI TO MEASURED RESULT.

A COMPLETE COMPUTATIONAL PIPELINE FOR EXPLAINABLE 3D BRAIN-TUMOR ANALYSIS.

TRANSPARENT STEPS.
REPRODUCIBLE RESULTS.



TECHNICAL SPECIFICATIONS



MODEL

SegResNet · 3D CNN · MONAI
BraTS 2018 pretrained weights



INPUT

T1 / T1c / T2 / FLAIR
(aligned NIfTI volumes)



INFERENCE

96³ window · 25% overlap
batch size 1



OUTPUT

Segmentation · Probability · Grad-CAM
Volumes · Centroids

THREE CASES. THREE OUTCOMES. ONE IMPORTANT LESSON.

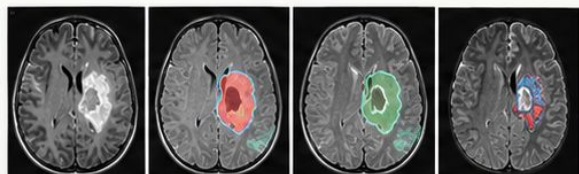
SAME PIPELINE. DIFFERENT REALITIES. A CLOSER LOOK AT MODEL BEHAVIOR.

CASE 00000

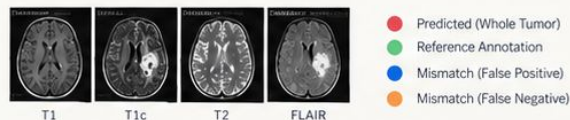
A WELL-SEGMENTED CASE

BrATS 2018
HGG (example)
256 × 256 × 155

Whole Tumor Dice: **0.918**



FLAIR (Axial) CNN Segmentation Reference Annotation Difference (|Pred - Ref|)

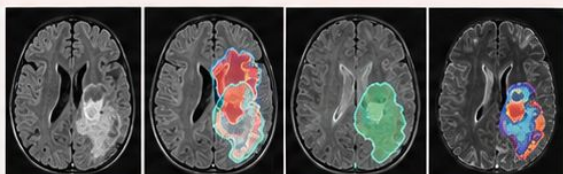


CASE 00002

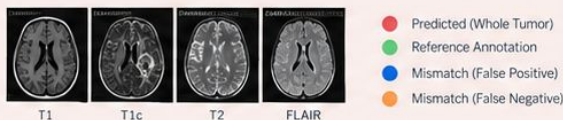
A CHALLENGING CASE

BrATS 2018
HGG (example)
256 × 256 × 155

Whole Tumor Dice: **0.458**



FLAIR (Axial) CNN Segmentation Reference Annotation Difference (|Pred - Ref|)

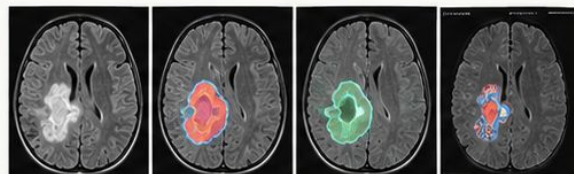


CASE 00003

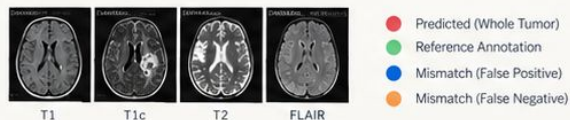
ANOTHER SUCCESSFUL CASE

BrATS 2018
LGG (example)
256 × 256 × 155

Whole Tumor Dice: **0.878**



FLAIR (Axial) CNN Segmentation Reference Annotation Difference (|Pred - Ref|)



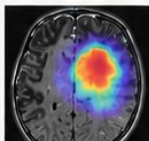
KEY TAKEAWAY

Probability ≠ accuracy.

The dashboard's mean tumor sigmoid is not calibrated diagnostic confidence and should not be interpreted as segmentation quality.

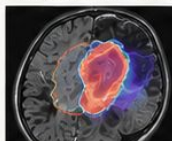
WHY THIS MATTERS

HIGH MODEL PROBABILITY



≠

LOWER SEGMENTATION AGREEMENT



High voxel-wise probability does not guarantee correct boundaries, and low probability does not always mean a failure.

OBSERVATIONS

- Model performs well on many cases, but struggles in some challenging scenarios.
- Probability and segmentation quality are related, but not equivalent.
- Visual inspection, reference comparison, and contextual analysis are essential.



Reference annotations make failure visible.

They enable quantitative evaluation, qualitative analysis, and a clearer understanding of model limitations.

MORE
CASES.
DEEPER
UNDERSTANDING.

FROM WORKING PROTOTYPE TO RESEARCH PLATFORM.

A FOUNDATION FOR MORE TRANSPARENT, REPRODUCIBLE AND IMPACTFUL BRAIN-TUMOR ANALYSIS.

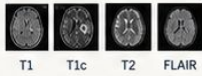
01 WHAT THIS PROJECT CONTRIBUTES

AN INTEGRATED RESEARCH WORKSPACE



Geometry-aware preprocessing

Standardize, align and normalize multi-modal 3D MRI (T1, T1c, T2, FLAIR).



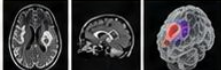
3D CNN segmentation

MONAI SegResNet for volumetric tumor segmentation.



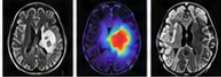
Interactive volumetric visualization

Multi-planar views and 3D rendering for intuitive exploration.



Regional Grad-CAM

Model attribution to highlight relevant image regions.



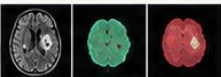
Physical-volume measurement

Voxel count $\times$ affine determinant for real-world units (mm^3).



Reference-mask comparison

Quantitative comparison with dataset annotations (e.g., Dice).



Deterministic + LLM-assisted reporting

Structured results with optional Gemini narrative for interpretability.



02 WHAT IT DOES NOT CLAIM

IMPORTANT BOUNDARIES



NOT A DIAGNOSTIC DEVICE.



Pretrained on glioma data

Optimized for the BraTS 2018 distribution. May not generalize to all tumor types.



Not an all-brain-tumor detector

Designed for glioma segmentation, not general detection or classification of all brain abnormalities.



Not clinically validated

Research prototype, not evaluated in clinical settings.



Probability is not calibrated confidence

The model's sigmoid output indicates relative likelihood, not diagnostic certainty.



Current cases are not an independent benchmark

Evaluated on available datasets; further validation on independent, diverse cohorts is required.

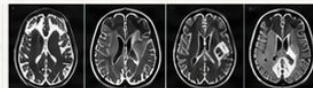
03 NEXT RESEARCH QUESTIONS

TOWARDS A MORE GENERAL AND IMPACTFUL SYSTEM



ROBUSTNESS

How does performance change across scanners, acquisitions and datasets?

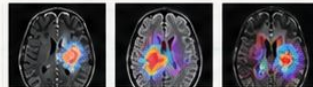


Different scanners | Different protocols | Real-world data



EXPLAINABILITY

Do attribution regions agree with clinically meaningful structures?

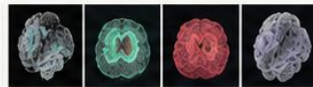


Model attribution | Anatomical context | Clinical relevance



GENERALIZATION

Can the workflow support additional tumor types and segmentation models?



Beyond glioma | Multiple models | Broader applicability



EVALUATION

Can larger independent cohorts establish reproducible performance?



Multi-center data | Diverse populations | Real-world validation

KEY CONCLUSION

The contribution is not only a segmentation model.
It is a **workflow** for turning model output into inspectable evidence.

MORE
TRANSPARENT
MODELS.
A CLEARER
TOMORROW.

SYSTEM.ONCO

RESEARCH
FOR A HEALTHIER
TOMORROW.